

What a world model’s uncertainty outputs actually report: an independent reproduction of the Robotic World Model

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Abstract

We independently reproduce the proprioceptive dynamics model of Li, Krause and Hutter (*Robotic World Model*, arXiv:2501.10100) and its uncertainty-aware follow-up (arXiv:2504.16680), building the model from scratch on CPU and verifying it against the released reference at the level of outputs, losses and gradients before training anything. The base paper’s central claim reproduces, and by a wide margin: over 3 training seeds, autoregressive training reaches normalised error 0.3582 ± 0.0283 at a 368-step horizon on held-out episodes against 1.6497 ± 0.2858 for teacher forcing, a factor of $4.61\times$, with the per-episode gap positive on all 10 of 10 episodes — an exact sign test at $p = 0.0020$. That verdict was fixed by a decision rule committed to git before the runs that tested it existed. Neither of the follow-up’s uncertainty outputs survives contact with a calibration measurement. The checkpoint emits a per-member **aleatoric** σ and an **epistemic** ensemble disagreement, and the method penalises rewards with the second while discarding the first. We measure both. At the 368-step deployment horizon the aleatoric σ is **20,669 $\times$ smaller than the realised error** on `n_independent = 20` trajectories, giving **0.02%** coverage at $\pm 1\sigma$ against a calibrated 68.3%, and we show analytically why: the state loss is squared error on a reparameterised sample with no log- σ term, minimised at $\sigma = 0$, with the bound term that should oppose this cancelling algebraically. Running the correction — the authors’ own unused `gaussian_nll` branch — fails differently rather than succeeding, at $10.9\times$ overconfidence. The epistemic term, the one the method actually consumes, is nearly three orders of magnitude better and still **34.4 $\times$ overconfident at the deployment horizon**, with 3.59% coverage where a calibrated Gaussian gives 68.3% (`n_independent = 20`). Two results run the other way, and both are tests the original work did not run. **Ensemble disagreement beats a trivial baseline.** Against the forecast step index — a counter, free, requiring no ensemble — disagreement correlates $+0.605$ [$+0.545, +0.694$] with realised error where the counter reaches $+0.269$ [$+0.106, +0.431$], and partialling the counter out changes disagreement’s correlation by $+0.010$. The follow-up’s trust-metric claim survives adversarial testing against a real alternative. **And the interval is repairable.** A per-horizon multiplier, fitted on one held-out episode and scored on the other, brings coverage within 10 points of target on 10 of 10 held-out cells, where a constant multiplier manages 2. Measuring all four models we trained or scored puts the failure precisely. The teacher-forced arm has the most input-dependent σ ($15.6\times$ the autoregressive arm’s) and ranks its own errors on 45 of 45 dimensions, and is still $315\times$ overconfident. That count is *not* the strong evidence an independent-trials test makes it look like: the 45 state dimensions are physically coupled and share a forecast-depth trend, and a permutation test over whole trajectories moves its P-value from $5.68e-14$ to 0.2609. No dimension count in this paper survives multiplicity correction under that test. These models can learn *which* predictions will be worse. They cannot learn *how wrong* they will be. A downstream user who needs a ranking may be served; one who needs an interval is not, under any of the four. We also report four defects in the released pipeline, and evidence that the released checkpoint’s variance state is not reachable from the released artifacts at the iteration count its author recalls — which he attributes to the repository having moved on between training and release. The gap is not marginal: the bound arithmetic implies roughly 158,000 iterations against the 5,000

recalled, a factor of about $32\times$ still unexplained (§7). And we report six retractions of our own numbered claims, plus two more that withdraw framings rather than numbers — that one of our own pre-registrations was pre-registered at all, and that the per-dimension counts throughout §5 could be read as independent trials. Every number in this paper is generated from a file in `results/`; none is typed by hand.

1 Introduction

A world model that reports its own uncertainty is more useful than one that does not, and the uncertainty-aware Robotic World Model reports one. This paper asks what that number means.

We came to the question sideways. Our aim was an ordinary reproduction: rebuild the proprioceptive dynamics model from scratch, check it against the released implementation, and see whether the paper’s central training claim holds. It does. But the same rebuild made a second question cheap to ask, because we had a from-scratch model, the released checkpoint, and a harness that could score both: *is the predicted σ calibrated?* Neither of the two the checkpoint emits is — the per-member σ by three to four orders of magnitude, the ensemble disagreement the method actually uses by one to two — and for the first of them the reason is structural rather than incidental.

This is a reproduction paper, and we mean the term in its stronger sense: the contribution is not that the numbers came out the same, but what systematically re-measuring the method reveals about where it is robust and where it is not. §9 collects the lessons in a form a practitioner can apply without reading the rest. Three things distinguish the work from a re-run of the authors’ code.

We rebuilt rather than imported. The forward pass, the loss and the training step are written from scratch and then checked against the reference: outputs match bitwise, and losses and gradients match to 0.000e+00 across 7 loss terms and 106 parameter tensors before any training begins (Appendix A). A discrepancy found later is therefore a property of the method, not of our wiring.

Decision rules were committed before the data. The verdicts below were fixed in advance, in git, with timestamps a reader can check (§8, Figure 4). One of them returned "cannot be settled" and we report that too.

We retract our own findings when they fail. Six claims in this work are withdrawn on evidence this project produced, and the retractions are kept in the record rather than deleted. One of them concerns the pre-registration discipline itself.

2 Setup

Data. The released dataset is 10,000 rows of ANYmal D proprioceptive state and policy actions at 50 Hz. It is not one recording: it is ten concatenated 20-second episodes, and its termination column is identically zero, so nothing in the file marks the boundaries. The reference window builder therefore marks all 9,961 windows valid, including 352 that splice one episode’s end onto the next one’s start. The usable, episode-respecting count is 9,609 — 10,000 rows, less 39 that cannot start a full window, less 352 that cross a boundary. The contamination rate is 3.53%.

Model. A GRU-based ensemble predicting the next proprioceptive state, with a mean head and a bounded log- σ head, plus auxiliary heads for contact and termination. The paper describes two loss terms; the implementation has 7.

Evaluation. Two arenas, held separate throughout: *out-of-sample*, the two episodes withheld from training, and *in-sample*, the eight used for it. We report both, because the released evaluation draws its trajectories from training data and the distinction is invisible in the original.

Effective sample size. Trajectory count is not sample size. Two 400-step trajectories whose spans overlap are not independent evidence, and the out-of-sample arena contains only 4 mutually non-overlapping 400-

step trajectories. Every interval in this paper is a bootstrap over independent trajectories, and every table reports that count.

3 What the original papers claim, and which claims we test

A reproduction that does not say what it left alone invites the reader to assume it tested everything. It did not.

Claim	Test
“The sign test, which does not depend on n. At $h = 368$ the per-episode gap favours autoregressive training on 10 of 10 episodes — an exact two-sided binomial test, $p = 0.0020$. This is one test on ten paired episodes, it uses no bootstrap, and no multiplicity correction touches it. Unlike the per-dimension counts in §5, episodes are genuinely separable units, so a binomial null is admissible here.”	“Sign test, which does not depend on n. At $h = 368$ the per-episode gap favours autoregressive training on 10 of 10 episodes — an exact two-sided binomial test, $p = 0.0020$. This is one test on ten paired episodes, it uses no bootstrap, and no multiplicity correction touches it. Unlike the per-dimension counts in §5, episodes are genuinely separable units, so a binomial null is admissible here.”

4 The base paper’s central claim reproduces

Claim under test. Training the dynamics model on its own autoregressive rollouts beats training it with teacher forcing, at deployment horizons.

Rule, committed in advance (commit `efc35b8`, and it names conditions rather than outcomes). Three conditions, all required: the out-of-sample gap at $h = 368$ excludes zero under a bootstrap over independent trajectories; the sign is consistent across episodes; and the effect survives at 10,000 iterations rather than only at the paper’s 2,500.

Result. Every condition holds. We give the evidence in order of how little it depends on the small held-out sample.

The sign test, which does not depend on n . At $h = 368$ the per-episode gap favours autoregressive training on **10 of 10** episodes — an exact two-sided binomial test, $p = 0.0020$. This is one test on ten paired episodes, it uses no bootstrap, and no multiplicity correction touches it. Unlike the per-dimension counts in §5, episodes are genuinely separable units, so a binomial null is admissible here.

Its scope needs stating plainly, because the arena labels invite a stronger reading than it supports. 8 of the 10 episodes are training data for *both* arms. The test is therefore a valid **paired** comparison — the two arms saw identical data, so any episode-level difference is attributable to the training rule and not to what either model had memorised — but it is not ten out-of-sample episodes, and it does not measure generalisation. The out-of-sample effect size below carries that burden, on 4 independent trajectories.

The in-sample arena, where the sample is larger. The same comparison on the eight training episodes has 16 independent 400-step trajectories against the held-out arena’s 4 — $4\times$ more — and gives the same direction at every horizon and checkpoint.

The out-of-sample effect size, over 3 seeds. Autoregressive training reaches **0.3582 ± 0.0283** against teacher forcing’s **1.6497 ± 0.2858** (standard deviation over seeds, `ddof=1`) — a factor of **$4.61\times$** .

Seed spread is not symmetric between the arms and that is worth stating: Arm A ranges 0.3341–0.3894 across seeds (7.9% relative), Arm B 1.4241–1.9710 (17.3%). Teacher forcing is more than twice as variable across seeds as autoregressive training at this horizon, so a single-seed comparison of these two arms is unreliable in a way a reader should know about. An earlier draft of this paper quoted the single-seed figures 0.3509 and 1.5540; those came from the seed that happened to be favourable to Arm A and unfavourable to Arm B, and the three-seed ratio is $4.61\times$ rather than $4.4\times$.

For a single seed the bootstrap over trajectories gives 95% interval $[0.56, 2.05]$ on $n = 4$ independent trajectories. **That interval should not be read as an ordinary one:** four trajectories admit 256 distinct resamples, so any bootstrap tail is quantised to steps of 0.39%, and the interval is coarse by construction. It is offered as corroboration of the sign test, not as the primary evidence.

Against a baseline, because neither number means anything without one. The hold-last floor — predicting that nothing changes — scores **0.9930** in the same cell. Autoregressive training beats it by **$2.8\times$** . **Teacher forcing is $1.56\times$ worse than assuming nothing changes at all**, which is the sharper statement of what

exposure bias costs here: the arm that reaches a lower training loss ends up predicting the future worse than a model that makes no prediction.

What does not hold, and we say so. At $h = 8$ — the horizon the model is trained on — the same comparison out-of-sample gives a gap of 0.008 whose interval includes zero. The advantage is a long-horizon phenomenon. An earlier rule of ours, anchored at $h = 8$, returned "cannot be settled"; anchoring a rule to the horizon the claim is actually about was a correction we had to make in advance of the runs, not after them (§8).

The pattern is consistent across the design. Under the correct cluster bootstrap, the out-of-sample gap excludes zero in **4 of 4** long-horizon cells — both trajectory lengths crossed with both checkpoints — and in **0 of 4** at $h = 8$. These figures are relative-L1; the nRMSE aggregation is reported separately and does not change the direction.

Multiplicity. Those 4 cells sit in a family of 8 out-of-sample comparisons, so we state the correction rather than leaving it to a reader. All 4 of 4 still exclude zero at a Bonferroni level of 0.05/8, and Holm–Bonferroni rejects **4 of 4**. The sign test above is unaffected either way.

5 Neither of the checkpoint’s uncertainty outputs is usable as an interval

5.1 Which quantity the method actually uses

The released checkpoint emits **two** uncertainty quantities, and the method consumes only one of them. This has to be settled before any calibration number means anything.

`system_dynamics.py:125` computes an **aleatoric** term, the mean over ensemble members of each member’s predicted σ . `system_dynamics.py:126` computes an **epistemic** term, the standard deviation across the members’ mean predictions. In `envs/base.py:142` the aleatoric term is bound to a local variable that is never read again; the epistemic term is stored, returned to the policy loop at :158, and applied at :166 as a reward penalty with weight -1.0 .

The paper agrees with its code. arXiv:2504.16680 Eq. 4 defines the penalised quantity as $u = \text{Var}_b[\mu_b]$, the variance across ensemble members, and Eq. 5 applies it as $\tilde{r} = r - \lambda u$. The per-member predicted variance enters the training objective and nothing downstream.

Eq. 4 specifies a **variance** while `system_dynamics.py:126` computes a **standard deviation**, which with $\lambda = 1$ differ by a square. We asked, and the first author confirms the code is operative: the penalty is applied to the standard deviation as intended, and Eq. 4 is "more of a high-level explanation" (personal communication, 21 August 2026). We measure the code’s quantity throughout, which is now known to be the intended one.

The same correspondence confirms the discard directly: "the aleatoric term is not used in downstream training. It is reported in Fig. 3 (right) as an analysis of the model behavior." So what follows is not an implementation slip being reported back to its authors — it is the intended design, and the aleatoric head exists to shape training and be inspected rather than to be consumed.

So the aleatoric head — the one the state loss and the bound loss shape, and the one §5.3 explains — is computed on every imagination step and discarded. We report both quantities below. Our own arms are ensemble size 1, where the epistemic term is identically zero by construction, so the epistemic measurement is possible only on the released checkpoint.

What the follow-up does and does not claim, stated before we measure anything. It does not claim its uncertainty is a calibrated interval. §5.1 claims the epistemic term "closely follows the trend of the prediction error" and that this "justifies its role as a trust metric", and of the aleatoric term it observes only that it "remains low, reflecting small stochasticity in the environment". Our measurement **supports the first claim** — the epistemic ordering is real and strong. What follows is therefore not a refutation of a calibration claim nobody made. It is three things the papers do not address: that the aleatoric head

is discarded before use, that neither quantity is usable as a scale, and that the low aleatoric value has a different cause than the one offered.

5.2 The measurement

For each model we compute the mean predicted σ , the mean absolute realised error, and the fraction of realised errors falling inside $\pm 1\sigma$. A calibrated Gaussian puts 68.3% inside $\pm 1\sigma$.

model	mean error / mean σ , whole 368-step rollout	coverage at $\pm 1\sigma$, h=1	coverage at $\pm 1\sigma$, h=368
faithful Arm A (sampled MSE)	52.2×	11.67%	1.75%
corrected Arm A (<code>gaussian_nll</code>)	10.9×	42.78%	8.57%
teacher-forced Arm B	315×	12.96%	0.56%
released checkpoint	7,878×	0.56%	0.04%

On the aleatoric head every model is overconfident by between one and four orders of magnitude (Figure 1) — that is the quantity §5.1 shows the method discards.

The quantity the method does use is also uncalibrated. On the released 5-member checkpoint over all 10 episodes, `n_independent = 20` non-overlapping 400-step trajectories. We use all ten rather than the held-out pair here because the released checkpoint trained on all ten, so restricting it to two buys no independence and costs four fifths of the sample — the same argument this paper makes about that checkpoint elsewhere. The held-out-only version at `n_independent = 4` is in the supplementary material (`results/task_b2_epistemic.json`). The epistemic column agrees in direction with this one at all 5 of 5 horizons; the aleatoric column agrees at 4 of 5 — it flips sign at h=8, where both readings sit close enough to chance that the sign is not meaningful in either. Where the two tables differ materially we say so.

h	aleatoric err/ σ	epistemic err/ σ	epistemic $\pm 1\sigma$	epistemic $\pm 2\sigma$	dims r>0	mean r	permutation P
1	1,827×	8.3×	16.22%	30.11%	44/45	+0.662	0.0060
8	3,034×	15.1×	9.99%	19.76%	45/45	+0.427	0.0085
32	4,525×	22.6×	6.95%	13.68%	45/45	+0.426	0.0355
128	14,934×	34.2×	4.37%	8.75%	45/45	+0.338	0.3725
368	20,669×	34.4×	3.59%	7.19%	45/45	+0.298	0.0823

Epistemic is nearly three orders of magnitude better than aleatoric at the deployment horizon — 600×

 — and still wrong by **8.3×** at one step and **34.4×** at the deployment horizon, with $\pm 1\sigma$ coverage of 3.59% where a calibrated Gaussian gives 68.3%. **Total** uncertainty, $\sqrt{\text{aleatoric}^2 + \text{epistemic}^2}$, equals the epistemic value to four significant figures at every horizon, because the aleatoric term is too small to move it.

The larger sample changes one thing materially, and it is a correction to our own earlier reading. At `n_independent = 4` the epistemic ordering looked like chance at short horizon — 23 of 45 dimensions at h=1 — and we had described it as a long-horizon effect. At `n_independent = 20` it is 44 of 45 at h=1, with mean r = +0.662, the *strongest* mean correlation of any horizon. The in-sample permutation test says the same (§5.5). The short-horizon "chance" result was an artifact of four trajectories, not a property of the model, and we record it as such rather than keeping the more interesting-sounding horizon story.

The last column gives permutation P-values over whole trajectories, not binomial ones, computed on the same 20 trajectories as the counts beside them; §5.5 explains why a binomial null is inadmissible here and how far it was wrong. These are five tests on one family and none survives Holm–Bonferroni across the arena’s 25 cells — the smallest is faithful (mse) h=368 at 0.0042 against a threshold of 0.002. Read the column as a consistency check on direction, not as five independent findings.

The scalar penalty as actually applied — `means.std(0).sum(-1)` at `envs/base.py:166` — correlates **+0.605** with total absolute error over the rollout, 95% CI [+0.545, +0.694] from a bootstrap over whole trajectories, `n_independent = 20` (7,360 pooled trajectory-step points). An earlier draft quoted this correlation with neither an interval nor an n. The interval resamples whole trajectories, not trajectory-step pairs, which would narrow it by about the square root of the rollout length.

5.3 Why the aleatoric head collapses: the optimum is $\sigma = 0$

This subsection explains the aleatoric column and only that column. Ensemble disagreement is not shaped by the mechanism below, and why *it* is miscalibrated is not established here.

It also supplies the alternative explanation promised in §5.1. The follow-up reads the low aleatoric value as reflecting "small stochasticity in the environment". The observation is correct and the reading is not: σ is low because $\sigma = 0$ is the optimum of the loss that trains it, and it would be low on any dataset, stochastic or not.

The state loss is squared error on a *sample* drawn from the predicted Gaussian, not a likelihood:

$$\mathcal{L} = \mathbb{E}[(\mu + \sigma\varepsilon - y)^2] = (\mu - y)^2 + \sigma^2$$

which is minimised at $\sigma = 0$ for any μ . There is no $\log\sigma$ term to oppose it. The bound term that appears to oppose it does not, because `max_logstd` is not an independent parameter — it is constructed as `min_logstd + exp(log_delta_logstd)`, so

$$\overline{\log \sigma_{\max}} - \overline{\log \sigma_{\min}} = \overline{\exp(\log \Delta_{\log \sigma})}$$

and `min_logstd` cancels algebraically, taking no gradient from that term. The floor the interval closes onto therefore freezes while the interval closes: a one-way ratchet.

We predicted the collapse from this algebra before training, then observed it. Across all 21 runs the collapse is linear in iteration count and its rate is nearly identical (Figure 3a). Rates are fitted on 15 of those runs: the 6 10,000-iteration runs are excluded from the rate statistics because they continue seeds already counted at 2,500 and would double-weight them. Figure 3(a) shows all 21 runs; Figure 3(b) plots only the 15 the rate is fitted on, so the scatter and the quoted statistic describe the same set.

The 21 runs, so a reader can count them:

arm	iterations	objective	dataset	seeds	seed ids
Arm A	2,500	gaussian_nll	clean	3	0, 1, 2
Arm A	2,500	mse	clean	3	0, 1, 2
Arm A	2,500	mse	contaminated	3	0, 1, 2
Arm A	2,500	mse	duplicated	3	0, 1, 2
Arm A	10,000	mse	clean	3	0, 1, 2
Arm B	2,500	mse	clean	3	0, 1, 2
Arm B	10,000	mse	clean	3	0, 1, 2

evidence that the mechanism is the objective and not the optimiser, the data or the architecture.

Two different things are being explained here, and §5.5 separates them. *Magnitude collapse is objective-driven.* It occurs in all 12 sampled-MSE runs at a rate of -9.3986e-05 per iteration with a standard deviation of 6.8e-07 — **including the teacher-forced arm**, which shares the objective — and reverses to +3.2332e-05 in the 3 runs that change it. *Input-independence is not.* That varies by a factor of 15.6 between two arms trained under the same objective, so the objective cannot be what produces it.

5.4 The correction fails differently rather than succeeding

The reference contains an unused `gaussian_nll` branch. Running it reverses the collapse and improves the magnitude from $52.2\times$ to $10.9\times$ overconfident. It does not produce a usable estimate, and it destroys something the faithful arm had: the σ -versus-error ordering falls from 39/45 dimensions positively correlated to 21/45, which is chance. Under the trajectory permutation test of §5.5 those counts give $P = 0.0417$ and 1.0000 out of sample, 0.0232 and 0.8734 in sample. The faithful arm's ordering is the one result in this family that points the same way in both arenas; it is also the weakest effect of the three, and it does not survive multiplicity correction either.

5.5 The failure is one of magnitude; the ordering is weaker than it looks

Measuring the teacher-forced arm — which we had trained for §4, and which our own first three calibration tables omitted — sharpens the finding:

model	σ variation across inputs (CoV)	dims with $r(\sigma, \text{error}) > 0$	perm P, out-of-sample	perm P, in-sample
faithful Arm A	0.0076	39/45	0.0417	0.0232
corrected Arm A	0.0059	21/45	1.0000	0.8734
teacher-forced Arm B	0.1188	45/45	0.2609	0.5656
released checkpoint	0.0177	20/45	0.6957	0.9738

Arm B’s σ is $15.6\times$ more input-dependent than the faithful arm’s, and it has the largest mean correlation of the four ($r = 0.257$). It is still $315\times$ overconfident.

The P column above is not a binomial one, and an earlier draft of this paper was wrong to make it one. Converting a count of positive per-dimension correlations to a P-value against a fair-coin null assumes the 45 state dimensions are independent trials. They are not. Position, velocity and torque for the same joint are physically coupled, and base linear and angular velocity are coupled through the gait. More importantly, error grows with rollout depth in every trajectory, so *any* σ that also grows with depth correlates with *any* trajectory’s error — including one it was never paired with.

We therefore permute whole trajectories. The null pairs each trajectory’s σ with a different trajectory’s realised error, which leaves both marginal distributions and the entire cross-dimension dependence structure intact and destroys only the association under test. The correction is large, and it is largest exactly where we leaned hardest. The worst-affected cell is teacher-forced armB at $h=368$, in the in-sample arena. It moves from $5.68e-14$ to 0.5656 — a factor of about 10^{13} — because under a null that preserves the dependence, a random re-pairing already yields 43.8 of 45 dimensions positive on average. Observing 45 of 45 against that null is close to unremarkable. A fair coin, by contrast, centres the count at 22.5 of 45; the dependence-preserving null centres it between 5.4 and 43.8 depending on model, horizon and arena.

So σ *collapsing in magnitude* is objective-driven, and σ *becoming input-independent* is not. The teacher-forced arm collapses in magnitude exactly like the autoregressive ones — same objective, same rate — while retaining $15.6\times$ more variation across inputs. Input-independence is a property of the autoregressive arms and the released checkpoint, and input-dependence and correct ranking are both achievable without the interval becoming meaningful.

One candidate mechanism, stated as a hypothesis and not a result: autoregressive feedback narrows the input distribution toward the model’s own manifold, leaving a heteroscedastic head less variation to key on. We have not tested it.

The same pattern holds for the quantity the method uses, and this is where the correction bites hardest. At $h=128$ and $h=368$ the epistemic term correlates positively with realised error on **45 of 45** dimensions, matching the best aleatoric head here on the sign count, while being $39.7\times$ overconfident. **All figures in this paragraph are the held-out arena ($n_independent = 4$),** so that the epistemic term and the four aleatoric heads are compared on identical trajectories; §5.2 quotes $34.4\times$ for the same ratio at $n_independent = 20$, which is the figure the abstract and §12 use. It does not beat Arm B’s head on strength either: its mean correlation at $h=368$ is $+0.151$ against 0.257 . The two quantities rank comparably; neither is close to an interval. Under the permutation null that count gives $P = 0.0435$ out of sample and 0.0758 in sample, against $5.68e-14$ from the independent-trials test we should not have used. It still fails the horizon test the same way: σ grows $1.59\times$ from $h=1$ to $h=368$ while error grows $13.33\times$.

The horizon story we first told was backwards, and the larger arenas agree with each other against the smallest. At $n_independent = 4$ out of sample, the epistemic ordering looked strongest at long horizon (0.0417 at $h=128$, 0.0435 at $h=368$) and unremarkable at short (0.4348 at $h=1$). Both larger arenas invert that. In sample ($n_independent = 16$): 0.0060 at $h=1$, 0.0084 at $h=8$, against 0.3804 at $h=128$. Over all ten episodes ($n_independent = 20$): 0.0060 , 0.0085 and 0.3725 . Two independent arenas at four and five times the sample say the effect is strongest at *short* horizon.

The null means explain why, and the explanation is the same one that motivates §5.6. At long horizon the shared forecast-depth trend lifts the null to 41.6 of 45, so a count of 45 is close to what chance alone delivers; at short horizon the null sits near 15.4 and the same count is genuinely surprising. The out-of-sample arena is not wrong so much as blind: at 4 trajectories its smallest attainable P-value is 0.04167, so it cannot distinguish a strong effect from a marginal one at any horizon. We report the small arena’s numbers alongside because it is the only arena that is out-of-sample for our own arms, not because it is the better measurement.

Nothing here survives multiplicity correction, in any of the three arenas. Holm–Bonferroni over each arena’s 25 model $\times$ horizon cells at $\alpha = 0.05$ rejects 0 out of sample, 0 in sample and 0 over all ten episodes. Out of sample that is a property of the design rather than of the models: with 4 independent trajectories the smallest attainable P-value is 0.04167, which already exceeds the smallest Holm threshold 0.002, so no effect of any size could have been rejected there. In sample the miss is real — the smallest P in the family is teacher-forced armB h=128 at 0.0027 against a threshold of 0.002.

So the honest form of this section’s claim is narrower than the one we first wrote. **The magnitude failure is established and large; the ordering is directionally consistent across every model and horizon we measured, and is not established at conventional significance once the dependence between dimensions is respected.**

The failure is specifically magnitude calibration, in both components.

5.6 Ensemble disagreement beats the trivial baseline

The follow-up justifies ensemble disagreement as a trust metric on the grounds that it "closely follows the trend of the prediction error". Section 5.5 shows the per-dimension version of that claim is weaker than it looks. This section asks a different and, for a practitioner, more important question: **does disagreement beat something free?**

Error in an autoregressive rollout grows with depth. So the trivial competitor to any trust metric is the forecast step index — a counter. It needs no ensemble, no second forward pass and no model. If a counter ranks error as well as disagreement does, the ensemble is not earning its cost. Neither paper runs this comparison, so we do.

All three correlations below are on the scalar quantity the method actually applies — `means.std(0).sum(-1)` at `envs/base.py:166` — against total absolute error, over `n_independent = 20` trajectories, with 95% intervals from a bootstrap over whole trajectories.

h	r(step index, error)	r(disagreement, error)	partial r(disagreement, error · index)
8	+0.181 [+0.108, +0.349]	+0.738 [+0.537, +0.807]	+0.757 [+0.540, +0.822]
32	+0.174 [+0.131, +0.336]	+0.735 [+0.577, +0.823]	+0.756 [+0.604, +0.841]
128	+0.526 [+0.421, +0.643]	+0.671 [+0.604, +0.846]	+0.617 [+0.534, +0.812]
368	+0.269 [+0.106, +0.431]	+0.605 [+0.545, +0.694]	+0.596 [+0.526, +0.687]

(h=1 has a single forecast step, so the index is constant and its correlation undefined.)

Disagreement wins at every horizon tested, and the intervals do not overlap. The counter reaches +0.269 over the full rollout against disagreement’s +0.605. The index leads in 0 of 4 horizons.

The third column is the one that settles it. Partialling the step index out of both variables leaves disagreement’s correlation at +0.596, against +0.605 raw — a change of +0.010. Almost none of what disagreement knows is explained by knowing how deep into the rollout you are. It is carrying real information about *this* rollout, not a re-encoding of the clock.

We had expected the opposite, and the brief that commissioned this test named it as the most valuable missing experiment precisely because a null would have been damaging. It is not a null. **On this axis the follow-up’s claim survives adversarial testing against a real baseline**, and that is the strongest form of support this paper offers any claim of either original work. It coexists with §5.5 without contradiction:

the *scalar* the method applies tracks error well, while the *per-dimension* sign counts we had leaned on carry far less evidence than an independent-trials test suggested. The quantity is a usable ranking signal and is still not an interval.

5.7 One constant scalar does not fix it, but a per-horizon one does

If σ had the right shape and the wrong scale, a single multiplier would repair it, and the finding would be a units problem with a one-line remedy. We tested that. A scalar was fitted on **one** held-out episode and evaluated on the **other**, in both directions, so it is never fitted on its own test set.

Fitting at one step works at one step and nowhere else. On the epistemic term — the quantity the method uses — a scalar of 5.08–5.82 brings $h=1$ coverage to 63–74%, essentially calibrated against the 68.3% target, and leaves $h=368$ at 17–21%. On the aleatoric term a scalar of 593–611 gives 64–70% at $h=1$ and 11–15% at $h=368$. Fitting over the whole rollout instead drives one-step coverage to 100% — an interval wide enough to be vacuous where the model is accurate — while still falling short at the far end.

The reason is §5.8’s mechanism: a constant multiplier cannot track an error that grows while σ does not. So "right shape, wrong scale" is the charitable reading of these tables, and for a *constant* scale it does not survive.

A per-horizon scalar does work, and this is the one concrete remedy in this paper. Fitting one multiplier per horizon on one held-out episode and evaluating on the other, in both directions, so no multiplier is ever scored on the episode that produced it:

quantity	held-out cells within 10 points of 68.27%, per-horizon c	same, constant c	range of fitted c
aleatoric	10 / 10	2 / 10	592.6 – 7782 (13.1×)
epistemic	10 / 10	2 / 10	5.082 – 47.33 (9.31×)

Every held-out cell lands within 10 points of the 68.27% target for both quantities; the worst is $h=1$ on the epistemic term at 74.44%. The constant scalar manages 2 of 10, and those are the $h=1$ cells it was fitted at.

Two cautions a reader should apply. The per-horizon scalar has five free parameters against the constant one’s one, so it *must* fit better in sample — only the held-out column above is evidence, and that is the column reported. And the correction is a calibration patch, not a fix: it leaves the model’s σ carrying no more information than before and simply rescales it by how far ahead you are looking. It is nevertheless enough to make the interval mean what it says, which is what a downstream user needs, and it costs one lookup table.

So the accurate form of this section is: **a constant scalar does not repair the interval; a per-horizon one does, and transfers across episodes.**

5.8 The structural excuse does not survive

One could argue that a model trained on an 8-step horizon cannot be expected to report calibrated uncertainty about step 368. It cannot report it about step 8 either. Inside the trained horizon, σ is flat while error grows (Figure 2):

model	σ growth, step 1 $\rightarrow$ 8	error growth, step 1 $\rightarrow$ 8
faithful Arm A	0.9241×	3.49×
corrected Arm A	1.0003×	3.41×
teacher-forced Arm B	1.0096×	6.11×
released checkpoint	1.0007×	1.79×

The faithful arm’s σ *declines* (0.9241×) while its error grows 3.49×. The coverage collapse in Figure 1(b) is therefore driven entirely by growing error against a fixed σ .

6 Defects in the released pipeline

6.1 Ten unmarked episode boundaries. §2. The window builder reads a termination column that is identically zero, so it marks all 9,961 windows valid.

6.2 Training and evaluation disagree on action alignment, and evaluation is the broken one. Row t holds the action that *produced* state t . The training path pairs states and actions index-for-index, which is causally correct. The evaluation path feeds the action from $t-1$ to predict state t — stale by one step. Scored correctly the released checkpoint is materially better than its own released evaluation reports: nRMSE at $h = 368$ falls from 1.3228 under the released pairing to 0.7572 under the causal one, so the released evaluation overstates its own model’s error by 75%.

6.3 No held-out evaluation. Evaluation trajectories are drawn from training data. For the released checkpoint, trained on the entire file, no held-out measurement is possible at all.

6.4 What the spliced windows cost: nothing measurable. We trained a contaminated arm on 7,882 windows — the clean 7,687 plus 195 splices — and, because that confounds *content* with *count*, a duplication control adding the same 195 windows as exact copies of windows already present.

The arm’s contamination rate is 2.47%, against the reference pipeline’s 3.53%. It is deliberately lower: we splice only the 5 boundaries whose *both* sides are training episodes, because 4 of the 9 put held-out rows into training. That is a leakage problem rather than a physics one, and including it would have invalidated our own comparison. So this experiment measures the cost of training on physically impossible transitions, and not the reference’s full exposure.

Training loss over the final 250 iterations: duplication costs 0.90%, splicing costs 21.57%. The bootstrap interval on duplicated — clean is $[-0.0061, 0.0215]$, including zero. So the rise is caused by splice content, not by dataset size — a control we ran only because the first version of this finding inferred the mechanism without it.

In rollout, across 32 cells (two arenas $\times$ two trajectory lengths $\times$ two checkpoints $\times$ two horizons $\times$ two metrics), contamination hurts in 0 of 32 and helps in 9 (Figure 5a). The control is inert, differing from clean in 2 cells.

"Costs nothing" is the wrong summary, and we should not use it. Splicing raises training loss by 21.57% against duplication’s 0.90%, and improves rollout in 9 of 32 cells. Both are measured effects in opposite directions, which is the signature of regularisation: the spliced windows contain transitions the model cannot fit, it fits the rest less tightly as a result, and it rolls out slightly better. The defensible statement is that **at this rate the splices do not harm rollout and appear to help slightly**, not that they cost nothing. **The unmarked boundaries remain a real defect on leakage grounds; what is now measured is that the physically-impossible-transition component costs nothing detectable at this rate.**

7 The released checkpoint’s variance state is unreachable at the stated iteration counts

The collapse rate is a clock. Fitting it across our runs and extrapolating to the released checkpoint’s σ state implies **153,270** optimisation steps at the configured learning rate. The refit from our 10,000-iteration runs gives 158,319 and 158,003, spreading 3.3% across the three fits — a linear extrapolation validated over a fourfold extension.

The released configuration says 500 iterations. The paper says 2,500. The checkpoint is tagged 5,000. A second, independent parameter on a slower gradient path implies the same order. And under `gaussian_nll` the implied count is *negative*, which identifies the branch the checkpoint was trained with.

What this extrapolation assumes, and what would falsify it. It assumes constant-rate Adam at the configured learning rate from the released initialisation. Five things would break it, and they are not equally plausible:

assumption	if violated	ruled out by the second parameter?
no learning-rate schedule	a decaying schedule inflates the implied count; a warm-up deflates it	partly — <code>min_logstd</code> and <code>log_delta_logstd</code> travel at rates differing by about 5×, and a uniform schedule scales both, so a schedule alone cannot reconcile them without also changing their ratio
<code>log_delta_logstd</code> initialised as released	a different initialisation moves the origin of the fit and rescales the count linearly	no — this is the weakest point of the argument
no warm start from an earlier checkpoint	a warm start makes the count a lower bound on total optimisation, not an estimate of one run	no
no gradient clipping in this path	clipping would slow the collapse and inflate the implied count	partly — the reference does not clip in the world-model path (X.08), so this is ruled out by source rather than by measurement
bound-loss weight as configured	a different weight scales the rate directly	partly — same ratio argument as the schedule

So the defensible claim is narrower than "cannot have come from the released recipe": **no constant-rate run from the released initialisation at the configured learning rate reaches this checkpoint's variance state in 500, 2,500 or 5,000 iterations.** A warm start or a different initialisation would explain the gap without any inconsistency, and we cannot exclude either.

What the author says. We wrote to the first author on 21 August 2026 asking exactly this. He replied the same day: the released `max_iterations: 500` is "a typo"; his recollection is 5,000 iterations, "as I always did"; he does not recall how the checkpoint was obtained; and — the part that matters most — "the checkpoint was released after a few iterations of the repo than the setup I used for the submission."

That last point reframes this section. The extrapolation above assumes the *released* initialisation and the *released* learning rate. If the repository drifted between the training run and the release, those are not necessarily the values that produced the checkpoint — and a changed `log_delta_logstd` initialisation is precisely the assumption the table above cannot rule out.

So the finding is not that the release is internally inconsistent. It is that **the released artifacts do not reproduce the released checkpoint's variance state, and the author's account is that the released repository is not the one that trained it.** That is a documentation gap between a release and a run — common, worth recording, and much less interesting than an inconsistency. We report the arithmetic because it is what let us detect the gap at all, not as a charge against the work.

8 Method

An append-only ledger. Every claim in this work has a permanent identifier, an evidence class (source, data, run, external, inference) and a status, in `FINDINGS_LEDGER.md` (171 entries). Claims are never edited in place. A claim that turns out to be wrong is marked superseded, with a pointer to what replaced it, and kept.

Pre-registration, and one failure of it. Decision rules were committed to git before the data that tested them — with one exception, which we report below. Figure 4 shows the lead time for each, computed from commit timestamps: the A/B rule by 1.3 hours, the flip-pattern rule by 4.8 hours, the difficulty-bias rule by 5 minutes, the long-horizon rule by 2 minutes.

The fifth bar is negative. The rule for the duplication control (§6.4) was stated in conversation before the runs but reached git **2.9 hours after the runs finished**, and we found this only by auditing our own git log. The measurement stands — the arm was built and run without reference to its outcome — but the claim that it was pre-registered does not, and we withdraw it. We report it because a discipline that is only checked when it succeeds is not a discipline.

Six retractions on our own evidence, out of 15 superseded claims kept in the record. In order: a premise about forecast decay that turned out not to exist in the code; a framing of the released checkpoint as "clearly informative" that rested on an $n=10$ estimate we ourselves showed to be biased low; an aggregation artifact that inverted a published-model comparison in our favour, withdrawn when the gating checks we had written refuted it; a per-dimension comparison that turned out to be unmatched; the claim that σ is input-independent "in all four models", made against a table holding three; and the phrase "the released checkpoint's uncertainty output", singular, when the checkpoint emits two and we had measured the one the method discards.

Two further retractions are **not** among those six, because they withdraw framings rather than numbers and are counted separately: the pre-registration claim above, and the inference from a count of positive per-dimension correlations to a binomial P-value, which assumed an independence the 45 state dimensions do not have (§5.5). The second was found by this paper's own submission review, and it is the one that cost

the most — it withdraws the strength of evidence behind what an earlier draft called the strongest result here.

A statistic that was resampling the wrong unit. Our bootstrap pooled three training seeds over a shared set of evaluation trajectories and resampled the pooled vector, while reporting the independent-trajectory count. Each trajectory appeared three times. Resampling trajectories correctly — carrying all seeds with each draw — widens intervals by a mean factor of $1.42\times$ (range 0.96–1.69) and changes 1 of 16 verdicts, in an $h = 8$ cell already recorded as unresolvable. Every long-horizon verdict survives. Both units are reported.

Reproducibility. `./reproduce.sh -quick -force` regenerates 19 artifact files and 4,804 numeric values from a clean clone, 4,804 of them bitwise identical (100.00%), 0 differing. Excluded and reported separately: 2,362 timing fields, and the 22 values in `results/step4_5_timing.json`. That file is the CPU budget — projected runtimes for configurations we did not run, peak resident memory, and the standard deviation across repeats — and every number in it measures the host rather than the model. It cannot reproduce bitwise on a different machine, or on the same machine under different load: one stage timed at 46.5 s idle took 109.7 s with training running concurrently.

9 Actionable lessons

Six things a practitioner can apply without reading the rest of this paper.

Use ensemble disagreement as a ranking signal; it earns its cost. Do not read it as a distance. The scalar the method applies correlates $+0.605$ [$+0.545$, $+0.694$] with realised error at $n_{\text{independent}} = 20$, beats the free alternative — the forecast step index — at every horizon we tested, and retains $+0.596$ once that index is partialled out (§5.6). That is a real signal, not a re-encoding of how far ahead you are looking. But it is $34.4\times$ too small to be an interval, and a risk gate or safety margin that reads σ as a distance is not supported at any horizon.

If you need the interval, rescale per horizon, not globally. One multiplier per forecast horizon, fitted on held-out data, brings coverage within 10 points of nominal on 10 of 10 held-out cells; a single global multiplier manages 2 (§5.7). The fitted multipliers span $9.31\times$ across horizons, which is precisely why one number cannot serve.

Do not convert per-dimension sign counts into P-values. State dimensions in a robot are physically coupled and share a forecast-depth trend, so an independent-trials null is badly wrong — in our tables by up to $10^{13}\times$ (§5.5). Permute whole trajectories instead. We shipped the binomial version in an earlier draft and it made our weakest evidence look like our strongest.

Count independent trajectories, not trajectories. Two 400-step windows that overlap at all are one piece of evidence, not two. The held-out arena here contains 4 independent 400-step trajectories however many windows are drawn from it, and that number — not the window count — bounds every long-horizon claim. Reporting an interval beside a trajectory count rather than an independent-trajectory count overstates precision, and resampling pooled seed $\times$ trajectory values instead of trajectories narrows intervals by a further $1.42\times$ (§8).

Anchor a decision rule to the horizon the claim is about. Our first pre-registered rule was anchored at $h = 8$, the training forecast horizon, and returned "cannot be settled". The claim was about deployment horizons. The rule was correct in form and pointed at the wrong regime, which is a failure mode that pre-registration does not protect against on its own.

Check that the implemented loss is the described loss before reproducing any number from it. The paper describes two loss terms; the implementation has 7. The predicted variance has an optimum at zero under the implemented one, which is why the released checkpoint's σ is $7,878\times$ smaller than its own error. Reading the loss took an afternoon and explained a result that would otherwise have looked like a training bug.

10 Broader impact

This is a reproduction of a dynamics model on public simulation data, and the reproduction itself carries no significant risk of harm: no new capability, no personal data, no deployment.

The finding does bear on safety, in one specific way worth stating. The method this paper examines uses its uncertainty estimate as a **trust metric** — a reward penalty that steers a policy away from states the model is unsure about. That use is supported by our measurements. But a downstream user who reads the same quantity as a *calibrated interval* — a safety margin, a confidence bound, a gate on when to hand control to a fallback controller — would be materially misled: at the deployment horizon the released checkpoint’s ensemble disagreement is $34.4\times$ smaller than the realised error, giving 3.59% coverage where 68.3% is expected. On hardware, a margin that is wrong by that factor is the difference between a conservative controller and one that believes it is safe when it is not.

We think that makes the finding worth publishing rather than the reverse, and it is the reason §5 reports coverage rather than only correlation.

11 Limitations

Effective sample size bounds every long-horizon claim. The out-of-sample arena has 4 independent 400-step trajectories. That is the binding constraint on §4, and no amount of trajectory oversampling changes it.

Ensemble size. Our main experiment runs at ensemble size 1 against the reference’s 5, for CPU budget, so our own arms have no epistemic component — it is identically zero by construction at ensemble size 1. The epistemic measurement in §5.2 is therefore made on the released checkpoint only, and we cannot say how ensemble disagreement would behave in a model we trained.

One dataset, one gait, one terrain. All commands are drawn from one bounded box and the gait is a single trot throughout. "Generalisation" here means across velocity commands, not across gaits or terrain.

The per-horizon recalibration is fitted and tested on two episodes only. §5.7’s remedy transfers across the two held-out episodes in both directions, which is the strongest test the released split allows, but two episodes is not a demonstration that the multipliers transfer to a new robot, gait or terrain. Treat the lookup table as a recipe to refit, not as constants to copy.

Two secondary analyses rest on a single training seed — the long-horizon trend fit and the per-dimension matched comparison, both computed on seed 1 alone. The headline A/B result is not among them: it is a three-seed mean with per-seed values reported (§4). This is recorded in the artifacts themselves.

We did not measure what the miscalibration costs. We show that the penalty the follow-up applies is miscalibrated as a scale — $34.4\times$ overconfident at the deployment horizon — but the only use the method makes of that quantity is to shape policy learning, and we did not train a policy. A miscalibrated scale that enters as a relative penalty across candidate actions may cost little, or may cost a great deal; our measurements cannot distinguish those. **The finding bounds what the quantity reports, not what it costs.** That distinction is easy to lose and we do not want a reader to take the ratio as a measure of harm.

The per-dimension ordering tests are underpowered at every sample size we can reach. Once the coupling between state dimensions is respected (§5.5), the out-of-sample arena’s 4 independent trajectories admit a smallest attainable P-value of 0.04167 — coarser than the multiplicity-corrected threshold 0.002, so that arena cannot reject at any effect size whatever. The larger arenas can reject and do not: over all ten episodes the smallest P in the family is 0.0042 against a threshold of 0.002. Resolving this needs more episodes than the released dataset contains, not a better test. Note the scope: this limits the *per-dimension* evidence. The aggregate scalar the method applies is separately and more strongly supported (§5.6), on the same trajectories, because it is one test rather than forty-five coupled ones.

We did not reproduce the policy-learning results of either paper. This is a dynamics-model reproduction only.

12 Conclusion

The Robotic World Model’s central training claim reproduces, and the margin is large. Neither uncertainty output of the follow-up that adds them reports what a reader would take it to report. The aleatoric σ is $7,878\times$ smaller than its own error, and the cause is that the objective’s optimum is $\sigma = 0$ with the term that should prevent this cancelling out of the gradient. The epistemic term the method actually penalises with is better by two orders of magnitude and still $34.4\times$ overconfident where it is used.

The more useful finding is asymmetric, and it cuts both ways. The scale failure is established and large — but it is repairable: a per-horizon multiplier, fitted on one held-out episode and scored on another, restores nominal coverage on 10 of 10 held-out cells where a global multiplier restores 2. And the ranking use the follow-up claims does survive a real test: against the forecast step index, a free baseline neither original paper ran, ensemble disagreement wins at every horizon and keeps $+0.596$ of its correlation once the index is partialled out. That is the one claim of either original work that this reproduction strengthens rather than qualifies.

What does not survive is the per-dimension form of the ordering evidence. Three of the five σ estimates we measured order their own errors better than chance in direction — the epistemic term on 45 of 45 dimensions at $h=368$, and the faithful and teacher-forced arms. The released checkpoint’s *aleatoric* head does the opposite, ranking backwards on every one of 45 dimensions, and the corrected arm sits at chance. And once the physical coupling between state dimensions is respected by permuting whole trajectories, no per-dimension count in this paper reaches significance after multiplicity correction. We report that rather than the independent-trials P-values an earlier draft carried, which were wrong by up to a factor of about 10^{13} on the cells we had cited as evidence. Neither quantity yields a usable interval. Uncertainty in this family of models should be read as a weak ordering at best, or fixed at the objective; it should not be read as a scale, and a ranking use deserves its own validation on the deployment distribution rather than trust inherited from here.

13 Data and code

The full repository — code, every artifact under `results/`, and `FINDINGS_LEDGER.md` with the complete claim record including the retractions — accompanies this submission as anonymised supplementary material, and will be released under a permanent archival identifier on acceptance. Neither upstream repository is redistributed; `setup.sh` fetches both at pinned commits and verifies two SHA-256 hashes.

The pre-registration argument in §8 rests on commit timestamps, and those are author-settable via `git commit -date`. That matters, because §8 is load-bearing. Two things address it. The supplementary material includes an anonymised `git log` covering every commit cited here, so the ordering in Figure 4 is checkable at review time. And **the repository was archived by Software Heritage on 21 August 2026**, before submission, under a permanent identifier whose visit timestamp is not author-controllable; the identifier resolves to a named repository and is therefore disclosed on acceptance rather than here.

What that archive establishes should be stated precisely, because it is easy to overclaim. It does **not** prove any individual commit date is genuine. It proves that the repository, with the whole pre-registration history in the form this paper cites, existed no later than that archival moment, as recorded by a third party with no interest in the claim — so nothing in the record can have been back-dated afterwards. That bounds §8 rather than proving it, and a reviewer should read it as such.

14 References

1. C. Li, A. Krause, M. Hutter. *Robotic World Model: A Neural Network Simulator for Robust Policy Optimization in Robotics*. arXiv:2501.10100, 2025.
2. C. Li, A. Krause, M. Hutter. *Uncertainty-Aware Robotic World Model Makes Offline Model-Based Reinforcement Learning Work on Real Robots*. arXiv:2504.16680, 2025.

A Verification chain

What every downstream number rests on. Each level was passed before the next was attempted.

level	claim	result
shapes	parameter counts match the reference	exact
wiring	inference outputs match the reference module	0.000e+00 , bitwise
indexing	the harness feeds the actions it claims	bitwise against the raw CSV
residual	the zero-delta model is the hold-last floor	1.192e-07
objective	losses and gradients match	0.000e+00 across 7 terms, 106 tensors
trainer	can memorise a single batch	1,506 $\times$ loss reduction

B Reproducing

```
./setup.sh # clone upstreams at pinned commits
python3.11 -m venv .venv && . .venv/bin/activate
pip install -r requirements.txt
./reproduce.sh --quick --force # everything except training
```

`-force` matters: a clean clone already contains each stage’s declared output, so without it every stage skips.

Runtime. Training stages are excluded by `-quick`, which is what makes the quick path practical. Training all 21 runs takes **32 hours** of recorded wall clock on two CPU cores: 20 hours for the 6 runs at 10,000 iterations and 12 for the remaining 15 at 2,500. The longest single run is 3.8 hours. An earlier version of this appendix said 22 hours; that figure predated the 6 ten-thousand-iteration runs added for the three-seed headline, and is corrected here from the `wall_clock_s` field of every run artifact rather than re-estimated.

C Figures





